

Advanced AI / ML / GenAI Project Roadmap

Overview

This roadmap outlines three key projects to build a strong, job-ready AI portfolio combining LLMs, Machine Learning, and Deep Learning.

Project 1: Document Intelligence RAG Assistant

Build a system that allows users to upload documents and query them using an LLM with retrieval.

- 1 Upload PDF/DOCX files
- 2 Extract and clean text
- 3 Chunk documents
- 4 Generate embeddings
- 5 Store vectors (FAISS/ChromaDB)
- 6 Retrieve relevant chunks
- 7 Generate answers with citations
- 8 Add logging and evaluation

Project 2: Customer Churn Prediction Pipeline

Develop a full machine learning pipeline for predicting customer churn.

- 1 Data cleaning and preprocessing
- 2 Feature engineering
- 3 Train multiple models (Logistic Regression, Random Forest, XGBoost)
- 4 Evaluate with metrics (F1, ROC-AUC)
- 5 Save best model
- 6 Create inference script or UI

Project 3: TensorFlow CNN Image Classifier

Build a deep learning model for image classification.

- 1 Load and preprocess image dataset
- 2 Build CNN using TensorFlow/Keras
- 3 Apply data augmentation
- 4 Train and evaluate model
- 5 Plot accuracy/loss
- 6 Generate predictions

Tech Stack Summary

- 1 Python
- 2 OpenAI API / LLM APIs
- 3 FAISS / ChromaDB
- 4 Streamlit
- 5 Pandas, NumPy, scikit-learn
- 6 TensorFlow / Keras
- 7 Matplotlib
- 8 Joblib

Execution Plan

- 1 Day 1: Build RAG system (core functionality)
- 2 Day 2: Build ML pipeline and CNN baseline
- 3 Focus on working demos, not perfection

Important Notes

- 1 Keep code modular (not just notebooks)
- 2 Add README with architecture explanation
- 3 Include screenshots and results
- 4 Focus on real-world applicability
- 5 Avoid overengineering UI